

AI Assisted Claims Data Challenge

Participant Exercise

Group name: _____

Assigned investigation: _____

Team members: _____

Suggested time: 30 minutes

Scenario

You are supporting an insurance operations team. Management wants a concise review of a synthetic claims dataset to identify operational patterns, data-quality concerns and areas that may require further review. Your job is to use AI as an analysis partner, verify its work in Excel and present a decision-ready conclusion.

Data safety and interpretation: Use only the supplied synthetic workbook and an approved AI tool. Never upload real customer, policy, medical or claims data to a consumer AI service. An unusual record is not evidence of fraud; it is only a reason to investigate further.

Your challenge

1 Inspect the workbook

Before prompting AI, spend three minutes reviewing the Claims Data sheet. Note:

- what one row represents;
- which fields are numerical, categorical, date-based or free text;
- any obvious missing, duplicated or potentially invalid values;
- two business questions the data could help answer.

2 Test a zero-shot prompt

Ask the AI the following with no example, method or output structure:

Analyse this Excel file.

Record what the response focused on, what it overlooked and any unsupported assumptions it made.

3 Improve the prompt

Now use this structured prompt. Compare the usefulness and reliability of the two responses.

You are a data analyst supporting an insurance operations team.

Analyse the Claims Data sheet in the attached synthetic workbook.

1. Describe the dataset, including the number of records and available fields.
2. Identify missing, duplicated or potentially invalid values.
3. Calculate: (a) total claim amount, (b) average claim amount, (c) average processing time and (d) number of claims by status.
4. Compare average claim amount and processing time by claim type, region and submission channel.
5. Identify up to five records that may require further review.
6. Recommend three useful charts.
7. Provide three operational observations supported by the data.

Present the results using clearly labelled tables. Do not invent missing information. Distinguish calculated findings from possible interpretations. State any assumptions and calculations used.

4 Complete your assigned investigation

Your facilitator will assign one question. Refine the prompt if needed. Ask the AI to show counts, calculations and assumptions—not only conclusions.

Investigation	Question
A Processing performance	Which claim types and regions have the longest average processing times? Include the number of records in every group and flag small groups that may produce unstable averages.
B Unusual claim amounts	Identify claim amounts that are unusually high relative to other claims of the same type. Explain the method used. Do not label a claim as fraudulent.
C Satisfaction	Examine the relationship between processing time and customer satisfaction. Create useful processing-time bands and explain why association does not prove causation.
D Submission channels	Compare Agent, Online and Branch submissions by claim count, average claim amount, average processing days, approval rate and average satisfaction.
E Management summary	Write a 150-word management briefing containing two evidence-based findings, one concern, one recommended action and one limitation of the analysis.

5 Verify the AI response in Excel

Check at least one important result independently with an Excel formula or PivotTable. If the figures differ, investigate the filters, treatment of blanks, grouping logic and calculation method.

Metric checked	AI result	Excel result	Match?	If different, why?

6 Create one chart

Create one chart in Excel that supports your conclusion. Be ready to explain:

- What business question does it answer?
- Why is this chart type appropriate?
- Could the chart mislead someone? If so, how would you improve it?

7 Prepare your group response

Give a 60–90 second presentation containing all five elements below:

- one verified finding;
- one data-quality issue or unusual record;
- one chart;
- one recommended action;
- one limitation or caution.

Prompting reflection

1. How did the structured prompt change the answer compared with the zero-shot prompt?
2. What did the AI overlook, calculate incorrectly or express with too much confidence?
3. Where did AI add value, and where was Excel alone sufficient?
4. Which parts still required human judgement?